

Algebraic & Geometric Methods in Statistics

Sampling distributions - an example

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Created for Math/Stat 561

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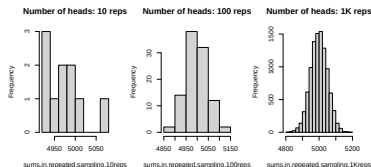
PART 1: Why do we care about distributions?

NOTE: This was in Lecture 2. Skip to Part 2 !!!!

- *Who cares* about model fitting and testing whether we have the correct model in the first place?
- Why do I have to *understand* a model?

Simulation of a coin toss

- Let x be the random variable recording the outcome of a coin toss:
 - $x_i = 0$ if we see Tail on the i -th trial (toss),
 - $x_i = 1$ if we see heads on the i -th trial.
- Fix $n = 10000$.
- $Y =$ the number of heads.
 - Is the number of heads supposed to be $n/2$? How far off is it? Does it vary? What does this mean?



- Sampling distribution of Y appears to have a mean around *the expected number of heads when a fair coin is tossed*, which is about $n/2$.
- The more times we repeat the experiment of n coin tosses, the closer Y gets to its expected value – this can be measured by looking at both the mean and the variance of Y .

Means of Y

4976.200

4997.500

4999.837

Vars of Y

2393.808

2393.808

2539.677

Question:

is it possible that something similar to this always happens?

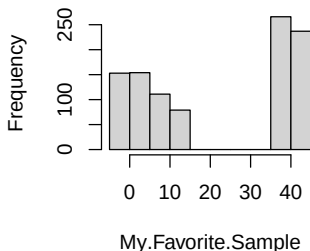
- As we will see, the sampling distribution of Y is approximately *normal* with mean equal to the expected value of X .
- In other words, the example above illustrates a known result—the **Central Limit Theorem**, one of the cornerstone results used in inference.
- You should already be familiar with it from your probability class.

Importance of sampling distributions

- **Sampling distributions tell a story** about the model behind the data (i.e., the probability distribution or population from which the data was sampled);
- they give a glimpse into how it was generated.

Example

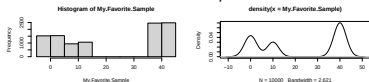
Histogram of My.Favorite.Sample



Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-2.8074	0.8944	37.6892	21.9786	39.9553	42.5403

Hmm...

- Is it strange to see “two bumps” in the histogram instead of one, as usual?
- Maybe the sample size is too small, we need to simulate more data?

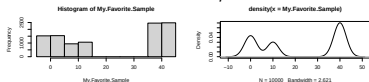


AhaMoment!

What do you see?

Hmm...

- Is it strange to see “two bumps” in the histogram instead of one, as usual?
- Maybe the sample size is too small, we need to simulate more data?



AhaMoment!

What do you see?

- This data is *not* being drawn from anything like a normal distribution.
- Consequently, knowing simply the mean and the variance ... is not enough to understand the data, that is, the data-generating mechanism behind it.

... Wait, what was that?!

This was an example of a **mixture** of normal distributions. We will see mixture distributions again in the course, soon. *(See also link in “License” page at the end of these slides, which includes source information.)*

PART 2: conditional independence models

Material is from chapter 4 of the textbook.

Independence

Two independent discrete random variables

Let $X \subset [n] := \{1, \dots, n\}$ and $Y \subset [m]$.

$$X \perp\!\!\!\perp Y \iff P(X = i, Y = j) = P(X = i)P(Y = j).$$

- In words, the **joint** probability factorizes as the product of the **marginal** probabilities. (Conditioning does not have an effect: recall definition of independent events from Lecture 2.)

Two independent continuous random variables

Let $X \subset \mathcal{X}$ and $Y \subset \mathcal{Y}$.

$$X \perp\!\!\!\perp Y \iff f_{X,Y}(x,y) = f_X(x)f_Y(y).$$

- In words, the **joint** density factorizes as the product of the **marginal** densities.

→ Extend definition of independence to sets of random variables. Example:
3 discrete random variables:

Independence

The distribution P is called *independent* if each probability is the product of the corresponding marginal probabilities:

$$P_{ijk} = P_{i++} \cdot P_{+j+} \cdot P_{++k}$$

Here, for instance,

$$P_{i++} = \text{Prob}(X = i) = \sum_{j=1}^b \sum_{k=1}^c P_{ijk}$$

The **independence model** has the parametric representation

$$\begin{aligned} \Theta &= \Delta_{a-1} \times \Delta_{b-1} \times \Delta_{c-1} \rightarrow \Delta = \Delta_{abc-1} \\ (\alpha, \beta, \gamma) &\mapsto (P_{ijk}) = (\alpha_i \beta_j \gamma_k) \end{aligned}$$

Figure 1: Source: Bernd Sturmfel's invitatio to Alg Stats lecture, SAMSI 2008

Conditional independence

Recall the definition of a conditional probability:

Definition 4.1.2. Let $A, B, C \subseteq [m]$ be pairwise disjoint. The random vector X_A is conditionally independent of X_B given X_C if and only if

$$f_{A \cup B|C}(x_A, x_B | x_C) = f_{A|C}(x_A | x_C) \cdot f_{B|C}(x_B | x_C)$$

for all x_A, x_B , and x_C . The notation $X_A \perp\!\!\!\perp X_B | X_C$ is used to denote that the random vector X satisfies the conditional independence statement that X_A is conditionally independent of X_B given X_C . This is often further abbreviated to $A \perp\!\!\!\perp B | C$.

Figure 2: Source: Algebraic statistics, Seth Sullivant, AMS-GSM book

Example: 3 r.v. and one CI statement

$X_A = X, X_B = Y, X_C = Z$ each vector is a single random variable.

$$f_{X|Y,Z}(x|y,z) = \frac{f_{X,Y|Z}(x,y|z)}{f_{Y|Z}(y|z)} = f_{X|Z}(x|z).$$

- Given Z , knowing X does not give any information about Y .
→ Check discussion after definition 4.1.2, page 73 of the book!

Marginal independence

Example 4.1.3 (Marginal independence). A statement of the form

$$X_A \perp\!\!\!\perp X_B := X_A \perp\!\!\!\perp X_B | X_\emptyset$$

is called a *marginal independence statement*, since it involves no conditioning. It corresponds to a density factorization

$$f_{A \cup B}(x_A, x_B) = f_A(x_A) f_B(x_B),$$

which should be recognizable as a familiar definition of independence of random variables, as we saw in Chapter 2.

Figure 3: Source: Algebraic statistics, Seth Sullivant, AMS-GSM book

- *Note:* Marginal independence is the same thing as independence of random variables: **factorization of joint as product of marginal densities**. We use the term ‘marginal’ when there are more random variables.

The mathematics behind conditional independence

- Suppose a random vector X satisfies a set of conditional independence statements. Which other conditional independence relations must the same random vector satisfy?
- There is *no finite set of axioms* from which *all* conditional independence relations can be deduced.
- There are some easy conditional independence implications, which are called **the conditional independence axioms** or **conditional independence rules**.

Conditional independence axioms

Proposition 4.1.4. *Let $A, B, C, D \subseteq [m]$ be pairwise disjoint subsets. Then*

- (i) (symmetry) $X_A \perp\!\!\!\perp X_B \mid X_C \implies X_B \perp\!\!\!\perp X_A \mid X_C$;
- (ii) (decomposition) $X_A \perp\!\!\!\perp X_{B \cup D} \mid X_C \implies X_A \perp\!\!\!\perp X_B \mid X_C$;
- (iii) (weak union) $X_A \perp\!\!\!\perp X_{B \cup D} \mid X_C \implies X_A \perp\!\!\!\perp X_B \mid X_{C \cup D}$;
- (iv) (contraction) $X_A \perp\!\!\!\perp X_B \mid X_{C \cup D}$ and $X_A \perp\!\!\!\perp X_D \mid X_C \implies X_A \perp\!\!\!\perp X_{B \cup D} \mid X_C$.

Figure 4: Source: Algebraic statistics, Seth Sullivant, AMS-GSM book

Work time!

Complete the **worksheet 1** handout in lecture.

Problem: complete the steps of the proof of the 4 CI axioms from this slide.

Additional example for hands-on work $\mapsto$ homework 2

Let

$$P^{(X_3=0)} := \begin{pmatrix} 0.05 & 0.15 \\ 0.075 & 0.225 \end{pmatrix}, P^{(X_3=1)} := \begin{pmatrix} 0.125 & 0.125 \\ 0.125 & 0.125 \end{pmatrix}.$$

Consider three binary random variables X_1, X_2, X_3 each taking values in the set $\{0, 1\}$ with joint probabilities $P(X_1 = i, X_2 = j, X_3 = 0) = P_{i,j}^{(X_3=0)}$ and $P(X_1 = i, X_2 = j, X_3 = 1) = P_{i,j}^{(X_3=1)}$.

- 1 Find the marginal distribution P_{X_1} of X_1 . (Recall that in the discrete case, integration is substituted by summation.)
- 2 Find the conditional distribution $P_{X_2, X_3 | X_1}$ of (X_2, X_3) given X_1 .
- 3 Is X_2 conditionally independent of X_3 given X_1 ?
- 4 Is X_1 conditionally independent of X_2 given X_3 ?

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This document is created for Math/Stat 561, Spring 2023-2026, at Illinois Tech.

Examples are drawn from other sources; for details see [this file with full references](#). That document also contains important questions you may wish to think about.

The worksheet is from Kaie Kubjas, handed out in our class with her permission.

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